Duke-NUS Medical School
Goh Cardiovascular Research Award (GCRA) from
The Gift of the Goh Foundation

Duke-NUS Medical School (Duke-NUS) is pleased to announce a Call for Proposals for funding awards made possible by the generous gift from the Goh Foundation.

Goh Cardiovascular Research Award Funding Programme
[November 2024 Call for Applications]

The Goh Cardiovascular Research Awards aim to support and enhance cardiovascular research impact at Duke-NUS and National Heart Centre Singapore (NHCS).

Eligibility:

1. Applicants must hold Tenure, Research, Educator, Practice and Clinical Track faculty appointments with Duke-NUS at the Assistant Professor level or above.
2. Applicants must be a full-time (i.e. >75% FTE) staff member of Duke-NUS or a SingHealth institution.
3. Awardees of the Goh Cardiovascular Research Award (GCRA) March 2024 at the point of application will not be eligible for the GCRA November 2024.

Proposal Review: A Goh Cardiovascular Research Award Review Committee will be appointed by the Dean to review the proposals. The funding priorities will be given to pilot study for novel hypothesis and new directions different from existing projects, addressing major scientific issues and unmet needs in cardiometabolic biology and medicine. Proposals focusing on multi-laboratory collaboration and teamwork, aimed at securing external team-based funding, will be prioritized for consideration.

Funding: The funding is up to S$100,000 for a period of one (1) year. No extension will be permitted. The total number of projects to be funded and the final funding amount will be determined by the availability of the fund and number of projects recommended by the Goh Cardiovascular Research Award Review Committee.

Reporting: Awardees are required to submit a Final Report to Duke-NUS within two (2) months after the project end date. Additionally, they are required to complete an annual Post-Research Grant Survey for two (2) years after the award period ends.

Closing Date: 06 December 2024, 5:00 pm

Please submit all materials in a single PDF copy to: t.geordi@duke-nus.edu.sg

The subject line of the email should read
"Goh Cardiovascular Research Award Proposal – November 2024".

1. TITLE OF PROJECT (Limit to 30 words)

IMPROVING THE CHAIN OF SURVIVAL: FORECASTING AND OPTIMIZING COMMUNITY FIRST RESPONSE FOR IMPROVED OUT-OF-HOSPITAL CARDIAC ARREST SURVIVAL OUTCOMES

2. EARLIEST PROJECT START DATE (if awarded):
1 April 2025

3. PRINCIPAL INVESTIGATOR (PI), CO-INVESTIGATORS AND COLLABORATORS:
*Please note that **only one** application per individual is allowed.*

Name of PI:	Lam Shao Wei Sean
Designation:	Deputy Director
Primary Institution / Department:	Singapore Health Services Pte Ltd, HSRC
Name of ACP / SRP:	Surgery
Office E-mail:	gmslasws@nus.edu.sg
Office Contact No:	65762617

Please list all co-investigators and collaborators of the research proposal.

S/N	Names of Co-Investigators and Collaborators	Designation	Institution
1	Fahad Javaid Siddiqui	**Assistant Professor**	**Pre-hospital & Emergency Research Centre, Duke-NUS Medical School, Singapore**
2	Ashish Kumar	**Assistant Manager**	**SingHealth Health Services Research Centre**

4. DETAILS OF RESEARCH PROPOSAL
Please provide a not more than 1-page (using Arial font size 10, single spaced) research plan on the scientific rationale, specific aims and brief description of the experimental design of the research proposal.

Scientific Rationale

- Out-of-Hospital Cardiac Arrest (OHCA) is a unique time-sensitive medical emergency that requires synchronized multi-agent coordinated action across the chain-of-survival[1] to ensure good outcomes.
- Outcomes are shown to be better when the agents involved in the care chain are capable and adequately equipped. The sooner interventions are delivered, the better the outcomes.
- Recent studies show that monthly OHCA incidence in Singapore increased from 2010 to 2019[2]. On the other hand, bystander Cardiopulmonary Resuscitation (CPR) prevalence has stagnated since 2017 and AED use has remained low despite significant increases in the prevalence of wall-mounted AEDs deployed across the country[3]. OHCA survivors continue to see a higher post-OHCA 3-year mortality rate relative to the general population [2].
- To improve the outcomes of OHCA patients, there is a clear need to identify and implement evidence-based care delivery strategies across the chain of survival, particularly at the initial stages of the OHCA incidents - in the community[4].

Specific Aims

Specific Aim 1: Assessment of OHCA community care demand vs community responder supply. We shall address the following research questions (RQs) under this aim:

- RQ 1.1 What is the forecasted incidence of OHCA till year 2040, based on Pan-Asian Resuscitation Outcomes Study (PAROS) data?
- RQ 1.2 What are the past and future projected distribution of OHCA incidence by location in Singapore?

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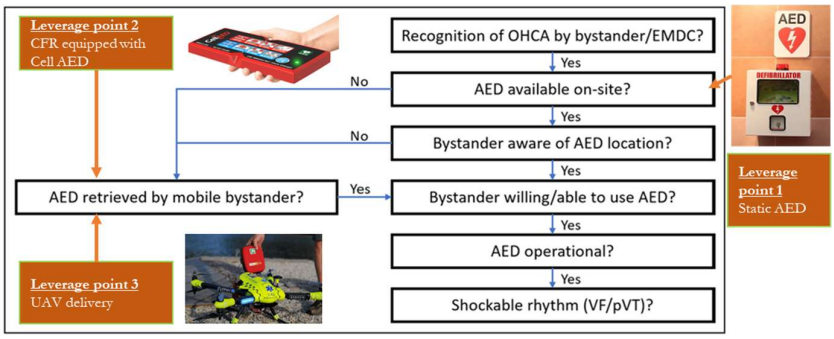
- RQ 1.3 What are the projections (or reasonable targets) for the prevalence of mobile community first responders equipped with AED?

Specific aim 2: Development of a decision support system to enhance AED-equipped community first responders (CFRs). Under this aim, the following RQs shall be addressed:

- RQ 2.1 Based on RQs 1.1-1.3 above, what is the cost-effective saturation of static AED deployments in Singapore?
- RQ 2.2: What is the optimal tradeoff between static and dynamic AEDs (CFRs equipped with portable AEDs and unmanned aerial vehicular delivery (UAV) of AEDs.?

This aim seeks to develop a decision support system that can evaluate near-optimal strategies (number and location) to deploy AEDs in both static and dynamic settings. The experimental design will evaluate alternative dynamic location-allocation strategies based on multi-stage mathematical optimization and simulation model. The following scenarios can be evaluated with this DDS:



- Base Scenario: Deployment of static AEDs following conventional practices (Leverage point 1 in Figure)
- Alternative Scenario: Base scenario enhanced with mobile AED-equipped CFRs (from both portable cell AEDs and UAV delivery) – Leverage points 2 and 3 in Figure.

Experimental Design

	Methodology/ tool	Main Inputs	Basic idea
WP1 OHCA Incidence forecast	• Gaussian Mixture Model for demand clustering	• PAROS data on OHCA incidence: location, patient demographics • Spatial demographics from public sources	• Plot locations of OHCA and derive clusters • Simulate (over 5-10 years) with demographic shifts (aging and new residential communities)
WP2 Optimization of last mile delivery	• Robust/Stochastic optimization • Scenario evaluation	• Demand clusters • Current locations of AEDs • Budgeted new AEDs (fixed and mobile)	• Formulate a mathematical simulation/optimization problem • Optimal set of AED locations, accounting for static/dynamic AEDs

Key features of the study

- The study team combines expertise in emergency medicine, public health, and systems engineering.
- The hybrid simulation-optimization approach will help to chart new directions (including mobile AED use) in OHCA management.

References

[1] C. D. Deakin, "The chain of survival: Not all links are equal," *Resuscitation*, vol. 126, pp. 80–82, May 2018, doi: 10.1016/j.resuscitation.2018.02.012.

[2] S. L. Lim *et al.*, "Temporal trends in out-of-hospital cardiac arrest with an initial non-shockable rhythm in Singapore," *Resusc. Plus*, vol. 16, p. 100473, Dec. 2023, doi: 10.1016/j.resplu.2023.100473.

[3] White, A.E., Shahidah, N., Asyikin, N., Liew, L.X., Pek, P.P., and Ong, M.E., "Singapore Out-of-Hospital Cardiac Arrest Report 2011-2020.," Unit for Pre-hospital Emergency Care, 2022.

[4] K. M. Simmons, S. M. McIsaac, and R. Ohle, "Impact of community-based interventions on out-of-hospital cardiac arrest outcomes: a systematic review and meta-analysis," *Sci. Rep.*, vol. 13, no. 1, p. 10231, Jun. 2023, doi: 10.1038/s41598-023-35735-y.

5. OUTCOME(S) OF RESEARCH PROPOSAL

In no more than 250 words (Arial font size 10, single spaced), please provide the proposed targeted outcome(s), including plans to secure new large extramural grants, to be involved in national initiatives, commercial development, etc. expected from the research proposal.

Rising OHCA incidence over the years coupled with the rapidly aging population will worsen the supply and demand parity for timely care of cardiac arrest patients. Technological advances offer some solutions, but these

are not only expensive to adopt but are complex to implement, demanding accurate assessment of the burden of disease, the shifting demand patterns and the effectiveness of solutions. OHCA chain of survival is significant in improving the outcomes of OHCA patients and effective and timely defibrillation is the starting point of this chain. The outputs of this project will provide a firm evidence foundation enabling us to design high-fidelity large-scale trials to evaluate the effective implementation of community first responders -based strategies in real-world settings in Singapore. The simulation-optimization based decision support platform (DSP) can be further utilized to evaluate other potential strategies to enhance community strategies to improve the timeliness of effective OHCA interventions. This proposal is based on robust local evidence which the Prehospital Emergency Research Centre (PERC) and Health Service Research Centre (HSRC) possess. DSP augmented with robust data are more likely to successfully attract extramural large funding from tech giants like Zoll, Laerdal, ZipLine, Phillips etc. The potential for adoption of any new technological solution as a standard of care will trigger local start-ups to develop custom solutions. The DSP is expected to save significant amount of financial and technical resources required to conduct numerous non-conclusive trials and help avoid unnecessary burden on the emergency care system.

6. BIOGRAPHICAL SKETCH AND LETTER OF SUPPORT

Applicant and all Co-Investigators to each provide a no more than 3-page NMRC-style biographical sketch (using template in ANNEX A). All collaborators to each provide a letter of support.

Name: Lam Shao Wei, Sean
Title: Dr
Office Mailing Address: SingHealth HSRC, Level 6, Academia, 20 College Road, Singapore 169856
Office Email: gmslasws@nus.edu.sg

Current Position
Deputy Director, Health Services Research Centre, Singapore Health Services Pte Ltd

Joint/Adjunct Positions
Associate Professor, Health Services and Systems Research, Duke-NUS Graduate Medical School, Feb 2016 - Now
Associate Professor, SingHealth Duke-NUS Global Health Institute, June 2021 - Now
Visiting Scientist, Changi General Hospital Feb 2016 - Now
Adjunct Faculty, Lee Kong Chian School of Business and SMU School of Information Systems Jan 2015 – Now
Percentage of time spent in Singapore every year: 100%
Academic Qualifications
PhD (2012) – Dept of Analytics and Operations; Dept of Industrial and Systems Engrg (NUS);
M.Eng (2002) (by research) - Department of Industrial and Systems Engrg (NUS);
B.Eng (1999) – Department of Mechanical Engineering (NUS)

Research Interests
I am active and contribute to Science in the fields of Healthcare Operations Research; Health Services Research; Medical Informatics and Healthcare Analytics. In my role as Deputy Director of Health Services Research Centre, we strive to bring data science, real world evidence, implementation science and health services modelling & evaluation to our healthcare system and alleviate the current problems.

Selected Publications in the last 5 years

1. Lam SW, Fang A.H.S., Koh M.S. *et al.* Development of a real-world database for asthma and COPD: The SingHealth-Duke-NUS-GSK COPD and Asthma Real-World Evidence (SDG-CARE) collaboration. *BMC Med Inform Decis Mak* **23**, 4 (2023). https://doi.org/10.1186/s12911-022-02071-6
2. SL Chan, AY Zhang, SSW Lam, V Rao, D Kanagalingam, HK Tan, PKH Chow, S Mathur. The impact of elective surgery postponement during COVID-19 on emergency bellwether procedures in a large tertiary centre in Singapore. Int J Qual Health Care 2024 Mar 21;36(1):mzae022. doi: 10.1093/intqhc/mzae022.
3. SL Chan, CZH Ho, NEE Khaing, E Ho, C Pong, JS Guan, C Chua, Z Li, T Lim, SSW Lam, LL Low, CH How. Frameworks for measuring population health: A scoping review. Plos One 2024 Feb 13;19(2):e0278434. doi: 10.1371/journal.pone.0278434. eCollection 2024.
4. LJ Davies, C Mathew, AR Pourghaderi, AXY Leong, DXH Chan, DLK Koh, AYH Tan, CYM Ong, J Ong, SSW Lam, SGK Ong. COVID-19 Pandemic Simulation Modelling in Anaesthesia Residency Training to Predict Delays and Workforce Deficiencies: A Case Study of the Singapore Residency Training Program. Cureus 2024 Jan 8;16(1):e51852. doi: 10.7759/cureus.51852. eCollection 2024 Jan.
5. SW Lam, Ge Y, Kumar A, Chen GZ, Pourghaderi A, Zaw MWW, Pierce KHC, Tan, HK, 2024. Regional surgical systems response to COVID-19 pandemic - Preliminary insights through longitudinal DEA. Forthcoming in Emrouznejad, A., Thanassoulis, E. and Toloo, M. (Eds.), *Advances in the Theory and Applications of Performance Measurement and Management*. Springer Nature.
6. SW Lam, Kumar, A , Guo, Y, Dorosan, M, Ong, MEH, Siddiqui, FJ. Simulation of Emergency Care Systems: A Taxonomy and Future Directions. Forthcoming in the proceedings of the Winter Simulation Conference 2024.
7. Ansah JP, Matchar DB, SW Lam, Low JG, Pourghaderi A, Siddiqui FJ, et al. The Effectiveness of Public Health Interventions Against COVID-19: Lessons from the Singapore Experience. *PLoS One*. 2021; 16(3). pp e0248742, doi: 10.1371/journal.pone.0248742.
8. EMP Fan, NHL Nguyen, SY Ang, F Aloweni, HQI Goh, LT Quek, TC Ayre, AR Pourghaderi, SW Lam, MEH Ong. Impact of COVID-19 on acute isolation bed capacity and nursing workforce requirements: A retrospective review. *Journal of Nursing Management* (2021). https://doi.org/10.1111/jonm.13260
9. N Liu, M Chee, C Niu, PP Pek, F Siddiqui, J Ansah, D Matchar, S Lam, H Abdullah, A Chan, R Malhotra, N Graves, M Koh, S Yoon, A Ho, D Ting, J Low, MEH Ong. BMC Med Res Methodol. 2020 Jul 2;20(1):177

10. Xie JG, GG Loke, M Sim, SW Lam. The Analytics of Bed Shortages: Coherent Metric, Prediction and Optimization. *Management Science*, (2019). (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3041878)
11. M Zhou, M Sim, SW Lam. Advance Admission Scheduling via Resource Satisficing. *Production and Operations Management*, 2019, (2019) [OPRE-2019-05-252] (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3394845).
12. AL Tan, CJ Chiew, SJ Wang, HAR Abdullah, SW Lam, MEH Ong, HK Tan, TH Wong. Scheduled elective surgery in an Academice Medical Centre: A Cohort Study. International Journal of Surgery, 66, 10.1016/j.ijsu.2019.04.009 (2019).

Research Grants

S/N	Project Title	Funding Amount	Funding Agency	Start Date	End Date
1	Citizen Data Science Programme for SingHealth Duke NUS AMC. AM/ACP-Designated Philanthropic Fund Award	$167,100	SingHealth Duke NUS AMC	1 April 2021	31 March 2024
2	Development of a Multi-Fidelity Simulation System for the Optimization of Operating Room Resources	$100,000	Duke - Duke NUS Collaborative Grant	1 October 2019	30 September 2021
3	Advancing Predictive and Prescriptive Analytics for the Optimisation of OT Resources	$50,000	SingHealth ACP Programme Funding, Clinical Innovation Support	1 April 2018	31 March 2019
4	Simulation Modeling and Optimization of Pre-hospital Emergency Care in Singapore	$100,000	MOH NMRC NIG Grant	2013	2015

Awards and Honors:

1. Commendation Medal, Pingat Kepujian, Prime Minister Office (https://www.pmo.gov.sg/National-Day-Awards/Information-On-Medals/The-Commendation-Medal-Pingat-Kepujian)
2. SingHealth Quality and Innovation Day. Best Poster in the Innovation category, Project Permelis, 2022, Singapore
3. SingHealth Quality and Innovation Day. Special Recognition Award, Project Permelis, 2022, Singapore
4. Singapore Healthcare Management Congress. Best Poster Award (Patient Experience), 2022, Project Permelis, 2022, Singapore
5. SingHealth Family – Target Zero Harm Award 2022 Team Award (nominee), Project Permelis, Singapore.
6. Singapore Healthcare Management Congress. Best Poster Award (Supply Chain Management), 2022, SHARP Project, 2022, Singapore
7. Singapore Healthcare Management Congress. Best Poster Award (Supply Chain Management), 2022, AED Placement Project, 2022, Singapore
8. Singapore Healthcare Management Congress. Best Poster Award (Communications), 2022, ODySSEy Project, 2022, Singapore
9. Singapore General Hospital Annual Scientific Meeting. Best Poster Award, COVID Predictive Model 2022, Singapore
10. SingHealth Duke NUS Research Team Award (2nd Prize) COVID-19 Virtual Disease Health System Outbreak Model for Singapore (CoViD-SOS), 2021, Singapore
11. Singapore Healthcare Quality Service Award (Special Mention) 2021. COVID-19 Virtual Disease Health System Outbreak Model for Singapore (CoViD-SOS) [National quality service award] (Singapore).
12. Singapore Healthcare Quality Service Award (Silver) 2019 [National quality service award] (Singapore).
13. Singapore Healthcare Management Conference Best Poster Presentation (Operations Section), 2019 (Singapore)
14. 15th IEEE Conference on Automation Science and Engineering (CASE), Best Healthcare Paper, 2019 (Canada).

15. Congress of the APSR (Asian Pacific Society of Respirology). Best Paper, Fukuchi Award, 2018 (Taiwan).
16. Surgical and Anaesthesia Congress 2017 (4-5 Aug 2017)– Best Poster Presentation (Surgery Category) (Singapore)
17. Singapore Healthcare Management Conference. Best Poster Presentation (Operations), 2017 (Singapore)
18. American Heart Association, Paul Dudley White International Scholar – International Science Team Award for Highest Ranked Abstract, 2017 (United States)
19. SingHealth YouShine Award, March 2017 – OT Dashboard Project for OT Optimization Task Force (Singapore)
20. SingHealth YouShine Award, Jul 2017–Analytics and Research Technologies for SingHealth (ARTS) (Singapore)
21. SingHealth Publish! Award, Medical Research, 2016 (Singapore)
22. INFORMS Healthcare Conference, Best Poster Presentation, 2015 (USA)
23. 6th International Conf on Information Management and Engineering Best Oral Presentation, 2014 (France)
24. Singapore Healthcare Management Conference Best Poster Presentation (Operations Section), 2013 (Singapore)
25. MTI Innovation Award from Ministry of Trade and Industry, 2012 (Singapore)
26. Most Cited Paper in Transportation Research Part C (Emerging Technologies) Paper 2012 (USA)
27. IIE Transactions Best Application Paper (Quality and Reliability Engineering Section), 2010 (USA)
28. Contributor to the book “Six Sigma: Advanced Tools for Black Belts and Master Black Belts”, Walter E. Masing Book Prize from the International Academy of Quality, 2007 (USA)

A. Career Long Service Awards

29. Singapore Health Services Long Service Award, 2021 (Singapore)

Name: Fahad Javaid Siddiqui
Title: Dr

Office Mailing Address: Pre-hospital Emergency Research Centre, Level 4, Duke-NUS Medical School, 8 College Road, Singapore 169856
Office Email: fahad.siddiqui@duke-nus.edu.sg

Current Position
Assistant Professor, Health Services & Systems Research, Duke-NUS Medical School.

Percentage of time spent in Singapore every year: 100%

Academic Qualifications

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Bolan Medical College, Balochistan University, Quetta, Pakistan	MBBS	Mar 1997	Medicine
The Aga Khan University (AKU), Karachi, Pakistan	MSc	Sept 2003	Epidemiology & Biostatistics
Curtin University of Technology, Perth, Australia	MSc	Feb 2008	Public Health

Research Interests
Evidence synthesis, Public Health, Health Services Research.

Publications in the last 2 years
Manuscripts forthcoming/ in progress

- Lam, S.S.W. , Kumar, A. , Guo, Y., Dorosan, M., Ong, M.E.H., **Siddiqui, F.J**. Simulation of Emergency Care Systems: A Taxonomy and Future Directions. Forthcoming in the proceedings of the Winter Simulation Conference 2024.
- **Siddiqui, F.J**. †, Kumar, A. †, Guo, Y., Ansah, J.P., Liu, Z., Ong, Malhotra, M., M.E.H., Lam, S.S.W. Long Range Forecasting for Emergency Care Systems in a Highly Dynamic Setting: A Singapore Case Study. Under review with *Journal of American College of Emergency Physicians*.

Publications (2023-24)

- Chen C, Lo CYZ, Ho MJC, Ng Y, Chan HCY, Wu WHK, Ong MEH, **Siddiqui FJ**. Global Sex Disparities in Bystander Cardiopulmonary Resuscitation After Out-of-Hospital. Cardiac Arrest: A Scoping Review. J Am Heart Assoc. 2024 Sep 17;13(18):e035794. doi: 10.1161/JAHA.124.035794. Epub 2024 Sep 9. PMID: 39248262.
- **Siddiqui FJ**, McNally B, Gräsner JT, Ho AFW, Tjelmeland I, Iwami T, Bray J, Wnent J, Lynn Lim S, Pin Pek P, Do Shin S, Ong MEH; GOHCAR Group. Towards advancing Out-of-Hospital cardiac arrest (OHCA) registries globally: Considerations from the global OHCA registry (GOHCAR) collaborative. Resusc Plus. 2024 Mar 22;18:100615. doi: 10.1016/j.resplu.2024.100615. PMID: 38549697; PMCID: PMC10973654.
- Wu J, **Siddiqui FJ**, Mak CCM, Chua ISY, Thangayah JR, Tan EXX, Seet HY, Rao AK, Tan HY, Mohamed A, Hartman M, Leong BS, Ong MEH, Mao DR. Assertive community treatment for high-utilizing alcohol misuse patients: a before-and-after cohort study protocol. BMC Health Serv Res. 2024 Feb 28;24(1):256. doi: 10.1186/s12913-023-10516-5. PMID: 38419049; PMCID: PMC10900701.
- Ong CA, Nadarajan GD, Fook-Chong S, Shahidah N, Arulanandam S, Ng YY, Chia MY, Tiah L, Mao DR, Ng WM, Leong BS, Doctor N, Ong ME, **Siddiqui FJ**. Increasing neurologically intact survival after out-of-hospital cardiac arrest among elderly: Singapore Experience. Resusc Plus. 2024 Feb 9;17:100573. doi: 10.1016/j.resplu.2024.100573. PMID: 38370311; PMCID: PMC10869923.
- **Siddiqui FJ**, Fook-Chong S, Shahidah N, Tan CK, Poh JY, Ng WM, Quah D, Ng YY, Leong BS, Ong ME. Technology activated community first responders in Singapore: Real-world care delivery & outcome trends.

Resusc Plus. 2023 Oct 13;16:100486. doi: 10.1016/j.resplu.2023.100486. PMID: 37859630; PMCID: PMC10582741.

- Tan YY, Chua ZX, Loo GH, Ong JSP, Lim JH, **Siddiqui FJ**, Graves N, Ho AFW. Risk of wound infection with use of sterile versus clean gloves in wound repair at the Emergency Department: A systematic review and meta-analysis. Injury. 2023 Nov;54(11):111020. doi: 10.1016/j.injury.2023.111020. Epub 2023 Sep 9. PMID: 37713965.
- Chee ML, Chee ML, Huang H, Mazzochi K, Taylor K, Wang H, Feng M, Ho AFW, **Siddiqui FJ**, Ong MEH, Liu N. Artificial intelligence and machine learning in prehospital emergency care: A scoping review. iScience. 2023 Jul 17;26(8):107407. doi: 10.1016/j.isci.2023.107407. PMID: 37609632; PMCID: PMC10440716.
- Goh JL, Pek PP, Fook-Chong SMC, Ho AFW, **Siddiqui FJ**, Leong BS, Mao DRH, Ng W, Tiah L, Chia MY, Tham LP, Shahidah N, Arulanandam S, Ong MEH; PAROS Clinical Research Network. Impact of time-to-compression on out-of-hospital cardiac arrest survival outcomes: A national registry study. Resuscitation. 2023 Sep;190:109917. doi: 10.1016/j.resuscitation.2023.109917. Epub 2023 Jul 26. PMID: 37506813.
- Lo CYZ, Fook-Chong S, Shahidah N, White AE, Tan CK, Ng YY, Tiah L, Chia MYC, Leong BSH, Mao DR, Ng WM, Doctor NE, Ong MEH, **Siddiqui FJ**. Does witness type affect the chance of receiving bystander CPR in out-of-hospital cardiac arrest? Resuscitation. 2023 Aug;189:109873. doi: 10.1016/j.resuscitation.2023.109873. Epub 2023 Jun 14. PMID: 37327852.
- Tan HS, Zeng Y, Qi Y, Sultana R, Tan CW, Sia AT, Sng BL, **Siddiqui FJ**. Automated mandatory bolus versus basal infusion for maintenance of epidural analgesia in labour. Cochrane Database Syst Rev. 2023 Jun 5;6(6):CD011344. Doi: 10.1002/14651858.CD011344.pub3. PMID: 37276327; PMCID: PMC10240562.
- Mak CCM, Mao DRH, **Siddiqui FJ**, Lim A, Davamoni-Thomas J, Tang JP, Bachik R, Ng CWL, Kandasami G, Lee C. A new paradigm in management of frequent attenders to emergency departments with severe alcohol use disorder-A pilot study for assertive community treatment in Singapore. Front Health Serv. 2022 Nov 4;2:1029455. doi: 10.3389/frhs.2022.1029455. PMID: 36925838; PMCID: PMC10012780.
- Chan SL, Lee JW, Ong MEH, **Siddiqui FJ**, Graves N, Ho AFW, Liu N. Implementation of Prediction Models in the Emergency Department from an Implementation Science Perspective-Determinants, Outcomes, and Real-World Impact: A Scoping Review. Ann Emerg Med. 2023 Jul;82(1):22-36. doi: 10.1016/j.annemergmed.2023.02.001. Epub 2023 Mar 14. PMID: 36925394.
- Ong MEH, **Siddiqui FJ**. Hidden disparities in the rising prevalence of bystander cardiopulmonary resuscitation. Nat Rev Cardiol. 2023 Mar;20(3):143-144. doi: 10.1038/s41569-022-00830-6. PMID: 36609603.
- Lim SL, Kumar L, Saffari SE, Shahidah N, Al-Araji R, Ng QX, Ho AFW, Arulanandam S, Leong BS, Liu N, **Siddiqui FJ**, McNally B, Ong MEH. Management of Out-of-Hospital Cardiac Arrest during COVID-19: A Tale of Two Cities. J Clin Med. 2022 Sep 1;11(17):5177. doi: 10.3390/jcm11175177. PMID: 36079106; PMCID: PMC9457434.
- Liu N, Liu M, Chen X, Ning Y, Lee JW, **Siddiqui FJ**, Saffari SE, Ho AFW, Shin SD, Ma MH, Tanaka H, Ong MEH; PAROS Clinical Research Network Investigators. Development and validation of an interpretable prehospital return of spontaneous circulation (P-ROSC) score for patients with out-of-hospital cardiac arrest using machine learning: A retrospective study. EClinicalMedicine. 2022 May 6;48:101422. doi: 10.1016/j.eclinm.2022.101422. PMID: 35706500; PMCID: PMC9096672.
- Lam SSW, Pourghaderi AR, Abdullah HR, Nguyen FNHL, **Siddiqui FJ**, Ansah JP, Low JG, Matchar DB, Ong MEH. An Agile Systems Modeling Framework for Bed Resource Planning During COVID-19 Pandemic in Singapore. Front Public Health. 2022 May 18;10:714092. doi: 10.3389/fpubh.2022.714092. PMID: 35664119; PMCID: PMC9157760.
- Tang XR, Zarisfi F, Pek PP, **Siddiqui FJ**, Malhotra R, Kwan YH, Ling T, Ho AFW, Ong MEH. Emergency department utilisation among older adults-Protocol for a systematic review of determinants and conceptual frameworks. PLoS One. 2022 Jun 3;17(6):e0265423. doi: 10.1371/journal.pone.0265423. PMID: 35661153; PMCID: PMC9166351.

† = joint first authors.

Name: Ashish Kumar
Title: Dr

Office Mailing Address: SingHealth HSRC, Level 6, Academia, 20 College Road, Singapore 169856
Office Email: ashish.kumar@duke-nus.edu.sg

Current Position
Assistant Manager, Singhealth Health Services Research Centre
Adjunct Senior Research Fellow, Health Services & Systems Research, Duke-NUS Medical School
Adjunct Lecturer, School of Computing and Information Systems, Singapore Management University (Jan-May 2025)

Percentage of time spent in Singapore every year: 100%

Academic Qualifications

Degree (year)	*Institution*	*Discipline*
Ph.D. (2011)	Indian Institute of Technology (IIT) Delhi	Management
Master's degree (1996)	Indian Institute of Management (IIM) Mumbai (then National Institute of Industrial Engineering, NITIE, Mumbai)	Industrial Engineering
Bachelor of Engineering (1992)	IIT Roorkee (then University of Roorkee)	Industrial Engineering

Research Interests
Health systems modelling; complex systems analysis with methods including agent based modelling, discrete event simulation, network analysis, and system dynamics.

Publications in the last 2 years
Manuscripts forthcoming/ in progress

- Lam, S.S.W. , Kumar, A. , Guo, Y., Dorosan, M., Ong, M.E.H., Siddiqui, F.J. Simulation of Emergency Care Systems: A Taxonomy and Future Directions. Forthcoming in the proceedings of the Winter Simulation Conference 2024.
- Siddiqui, F.J. †, Kumar, A. †, Guo, Y., Ansah, J.P., Liu, Z., Ong, Malhotra, M., M.E.H., Lam, S.S.W. Long Range Forecasting for Emergency Care Systems in a Highly Dynamic Setting: A Singapore Case Study. Under review with *Journal of American College of Emergency Physicians*.
- Kumar, A. †, Lam, S.S.W. †, Chan, S.L., Xu, Y., Yao, G., Tack, G.G.K, Khoon, H.K. Strategizing towards a Hospital of the Future: A Case Study on Collaborative Long-range Complex System Design. Accepted for International System Dynamics Conference 2024; manuscript under revision for journal submission.

† = joint first authors.

Publications (2023-24)

- Lam, S.S.W, Ge Y., Kumar A., Chen G.Z., Pourghaderi A., Zaw M.W.W, Pierce K.H.C, Tan, H.K, 2024 (Forthcoming). Regional surgical systems response to COVID-19 pandemic - Preliminary insights through longitudinal DEA. Forthcoming in Emrouznejad, A., Thanassoulis, E. and Toloo, M. (Eds.), *Advances in the Theory and Applications of Performance Measurement and Management*. Springer Nature.
- Kumar, A., Momaya, K. S., Liew, L., & Lam, S. S. W., 2024. Developing Human Capital for Competitiveness: A Study of Singapore's Artificial Intelligence Apprenticeship Program (AIAP) from a Systems Thinking Viewpoint. In Cho, D.S and Moon, H.C. (Eds.), *The Competitiveness of Nations 3: Emerging Technologies in the Fourth Industrial Revolution* (pp.219-246). World Scientific.
- Kumar, A., Liu, Z., Ansah, J.P., Ng, Y.Y., Leong, B.S.-H., Matchar, D.B., Ong, M.E.H., Siddiqui, F.J., 2023.

Viewing the Role of Alternate Care Service Pathways in the Emergency Care System through a Causal Loop Diagram Lens. *Systems* 11, 215.

- Matchar, D.B., Lai, W.X., Kumar, A., Ansah, J.P., Ng, Y.F., 2023. A Causal View of the Role and Potential Limitations of Capitation in Promoting Whole Health System Performance*. International journal of environmental research and public health* 20, 4581.

7. PROPOSED BUDGET

Awards are funded at a maximum of S$100,000 for a period of 1 year. Please prepare the budget carefully under each category. Funds cannot be utilised for salary support of the Principal Investigator. Up to S$5,000 can be budgeted for travel support for the Principal Investigator or research personnel who are working on the project. Budget must include GST, wherever applicable.

	Year 1	Total (S$)
EXPENDITURE ON MANPOWER (EOM)		
Analyst	$82,800	$82,800
Sub-total (S$)		$82,800
OTHER OPERATING EXPENDITURE (OOE)		
Publication	$5,000	$5,000
Conference travel	$5,000	$5,000
Intern services	$7,200	$7,200
Sub-total (S$)		$17,200
EQUIPMENT		
Sub-total (S$)		
Grand Total (S$)		**$100,000**

8. PLANS FOR EXPENDITURE OF REQUESTED FUNDS

In no more than 200 words (Arial font size 10, single spaced), please provide brief details of plans for the utilisation of the requested funds as well as the budget justification

We request for recruitment of intern for 6 months to assist with the analysis work required for project.
We aim to publish a manuscript and an abstract acceptance by the end of the project, hence, requesting for funds to travel for the conference as well as for publication fees.

We are also seeking funds for manpower assistance to lead the analysis of the project.

9. UNDERTAKING BY PRINCIPAL INVESTIGATOR:

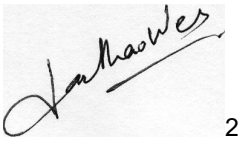
In submitting the proposal, the Principal Investigator undertakes to:

- Declare that all information is accurate and true.
- Declare that he / she meets the eligibility criteria of the GCRA programme.
- Submit supporting documents of ethics approval obtained from the relevant Institutional Review Board (IRB) and Animal Ethics Committee for studies involving human subjects and animal experimentation respectively, prior to initiation of the studies.
- Be actively engaged in the execution of the research and comply with all laws, rules and regulations pertaining to animal and human ethics, including the Singapore Good Clinical Practice guidelines.

Please check accordingly:

☒ I accept the above undertaking.

☐ I do not accept the above undertaking.



21/22/24

Sean Lam

Name of Principal Investigator **Signature and Date**

NOTE:

1. **Review Process**: Proposals will be reviewed and assessed by the GCRA Review Committee appointed by the Dean of Duke-NUS. The earliest start date for funded projects is 01 April 2025.

2. **Review Criteria**: Proposals will be evaluated based on the following criteria:
   - **(i) Scientific Merit** – What is the scientific merit and feasibility of the proposed studies? Priority will be given to charting new directions, forming new multi-disciplinary teams and developing / incorporating emerging new approaches.
   - **(ii) Principal Investigator** – Has the applicant demonstrated an aptitude and attitude towards excellence in science?
   - **(iii) Relevance and Potential Impact** – Does the research proposal address major scientific issues and unmet needs in cardiovascular biology and medicine?

ANNEX A

Instructions:
1. GCRA applicant and each Co-investigator to provide a no more than 3-page biographical sketch, using this NMRC format template.
2. Please indicate N.A. if the required information is not applicable.

- Name
- Title
- Office Mailing Address
- Office Email
- Office Contact No
- Current Position (Please provide full details, e.g. joint appointments; other academic appointments including those outside of Singapore)
- Percentage of time spent in Singapore every year (% declared should exclude **only** time spent overseas due to overseas appointment(s). Please indicate reason(s) if % declared is less than 100%.)
- Academic Qualifications (Indicate institution's name and year degree awarded)
- Research Interests
- Publications in the last 5 years (include only publications of direct relevance to study, stating impact factors where possible)
- Patents held (related or unrelated to the study)
- Scientific Awards
- Research Grants

Half page summary of research outcomes from all previous grants [e.g. publications (full papers only for the past 5 years and highlight papers that are relevant to the study), patents, awards, etc]

Personal Data Notice & Consent
For
Grants & Awards Managed and Administered by Duke-NUS Medical School

i. National University of Singapore ("NUS") through its Duke-NUS Medical School ("Duke-NUS") manages and administers grants and awards and the applications for the same. To do so, NUS will necessarily need to collect, use, disclose and/or process certain personal data or personal information submitted or otherwise made available to NUS by an individual in an application for a grant or award (the "**Applicant**").

ii. Such personal data will be collected, used, disclosed and/or processed by NUS for the purpose(s) of:

a) Administering, managing and/or processing the Applicant's application for a grant or award, sending the Applicant materials on the grantor awards applied for, and obligations thereunder, via postal mail, electronic mail, fax or voice calls;

b) Administering and/or managing the Applicant's relationship with NUS (including the mailing of correspondence, statements or notices to the Applicant, which could involve the disclosure of certain personal data about the Applicant to bring about delivery of the same, as well as on the external cover of envelopes / mail packages);

c) Carrying out screening activities (including background checks and/or obtaining of references and/or other information about the Applicant from the Applicant's previous education institute(s), employers) or other due diligence in accordance with legal or regulatory obligations or risk management procedures that may be required by law or that may have been put in place by NUS;

d) Responding to any enquiries by the Applicant;

e) Investigating fraud, misconduct, any unlawful action or omission relating to the Applicant's application for a grant, whether or not there is any suspicion of the aforementioned;

f) Responding to requests for information from public agencies, ministries, statutory boards or other similar authorities (including but not limited to the Immigration and Checkpoint Authority, Ministry of Defence, Ministry of Education and Ministry of Health) or non-government agencies authorized to carry out specific Government services or duties from time to time;

Where the Applicant is awarded the grant or award, the following purposes shall apply in addition to the aforementioned purposes set out in (b) to (f) which continue to apply until the grant or award comes to an end:

g) Administering and/or managing the successful Applicant's grant or award, which may include use and disclosure of personal data to related grant funding organisations, donors, external evaluators and/or external organisations for purposes of periodic reports as may be required under the grant or award conditions, and/or event invitations.

h) Maintaining campus safety and security of persons and property (including the use of CCTVs), where the successful Applicant is given access to NUS premises and facilities;

i) Invitation to participate in publicity materials, feature stories, events and other engagement activities relating to academic support, student support, financial assistance programs and

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merit-based awards made possible by donors, where the Applicant is a successful recipient, beneficiary and/or participant;

j) Carrying out market related or similar research and analysis for NUS' operational strategy and policy planning purposes;

k) Taking of photographs and/or videos (whether by NUS staff or third party photographers and/or videographers) during events or seminars organised by NUS or its affiliates for publicity purposes (including in NUS' publicity and communications materials).

l) Transferring and disclosing the successful Applicant's personal data to overseas partner institutions to review, manage, process the application and/or assess the Applicant's research. In such event NUS, through Duke-NUS will only share the personal data relevant and required to assess the research carried out under the grant or award.

iii. Such personal data of the Applicant may be disclosed by NUS to third party service providers or agents (whether in Singapore or elsewhere in the world where such service providers are sited) for one or more of the above Purposes, as such third party service providers or agents, if engaged by NUS, would be processing the Applicant's personal data for NUS, for one or more of the above Purposes.

iv. By completing and submitting the grant application form to Duke-NUS, the Applicant:

a) Declares and warrants that for any personal data of other individuals disclosed by the Applicant in connection with the grant application, the Applicant has, prior to disclosing such personal data in the grant application form to Duke-NUS, obtained the consent from the individuals whose personal data are being disclosed, to permit NUS through Duke-NUS to collect, use and disclose such personal data for purposes related to this application,

b) Consents to NUS collecting, using, disclosing and/or processing the Applicant's personal data for the Purposes as described above; and

c) Consents to NUS transferring the Applicant's personal data out of Singapore to NUS's third party service providers or agents and/or partner institutions for the Purposes as described above.

v. If the Applicant has any questions relating to NUS's collection, use and disclosure of the Applicant's personal data, the Applicant may contact the NUS Data Protection Office at dpo@nus.edu.sg or call 6776 2835 or such other person as NUS may designate, from time to time and informing the Applicant through letter, circular, notice or email.

vi. For the avoidance of doubt, in the event that Singapore personal data protection law permits an organization such as NUS to collect, use or disclose the Applicant's personal data without the Applicant's consent, such permission granted by the law shall continue to apply.

BEFORE YOU SUBMIT THE GRANT APPLICATION FORM TO DUKE-NUS, CAREFULLY READ THE ABOVE CONTENTS OF THIS DOCUMENT. BY SUBMITTING THE APPLICATION, YOU ACKNOWLEDGE THAT YOU HAVE READ AND FULLY UNDERSTAND THE CONTENTS OF THIS DOCUMENT AND YOU HEREBY GIVE YOUR CONSENT AS SET OUT ABOVE AS WELL AS CONSENT TO NUS COLLECTING, USING, DISCLOSING, PROCESSING AND/OR TRANSFERING OUT OF SINGAPORE, YOUR PERSONAL DATA FOR ONE OR MORE OF THE PURPOSES DESCRIBED IN THIS DOCUMENT.

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